

SF1811 Optimization

Lecture 13: Nonlinear optimization with nonlinear constraints

A note on optimality conditions, Part B

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Nonlinear optimization with nonlinear constraints

Main question:

When do linearized constraints describe the true local feasible geometry well enough for first-order optimality?

Today:

- Linearization of nonlinear constraints.
- Linearized feasible and descent directions.
- Why a constraint qualification is needed.
- The Mangasarian–Fromovitz constraint qualification.
- Karush–Kuhn–Tucker conditions.
- Sufficiency for convex problems.

The nonlinearly constrained problem

Consider

$$(P) \qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h_i(x) = 0, \quad i \in \mathcal{E}, \\ & g_i(x) \leq 0, \quad i \in \mathcal{I}, \end{array}$$

where f, h_i, and g_i are continuously differentiable.

At a feasible point $\hat{x}$, define the active inequality set

$$\mathcal{I}_a(\hat{x}) = \{i \in \mathcal{I} : g_i(\hat{x}) = 0\}.$$

This is the nonlinear counterpart of the linearly constrained problem from the preceding lecture.

The main difference is that the feasible set is now curved.

Linearization at a feasible point

Near a feasible point $\hat{x}$, write $x = \hat{x} + d$.

The first-order approximations are

$$h_i(\hat{x} + d) \approx h_i(\hat{x}) + \nabla h_i(\hat{x})d,$$

and

$$g_i(\hat{x} + d) \approx g_i(\hat{x}) + \nabla g_i(\hat{x})d.$$

Since $\hat{x}$ is feasible,

$$h_i(\hat{x}) = 0, \qquad i \in \mathcal{E},$$

and

$$g_i(\hat{x}) = 0, \qquad i \in \mathcal{I}_a(\hat{x}).$$

Thus the linearized active constraints are

$$\nabla h_i(\hat{x})d = 0, \qquad i \in \mathcal{E},$$
$$\nabla g_i(\hat{x})d \leq 0, \qquad i \in \mathcal{I}_a(\hat{x}).$$

The linearized feasible-direction problem

A vector d is a linearized feasible direction at $\hat{x}$ if

$$\nabla h_i(\hat{x})d = 0, \qquad i \in \mathcal{E},$$

and

$$\nabla g_i(\hat{x})d \leq 0, \qquad i \in \mathcal{I}_a(\hat{x}).$$

A linearized feasible direction is also a descent direction for the objective if

$$\nabla f(\hat{x})d < 0.$$

The search for such a direction is the linear program

$$\begin{aligned} \underset{d}{\text{minimize}} \quad & \nabla f(\hat{x})d \\ \text{subject to} \quad & \nabla h_i(\hat{x})d = 0, \quad i \in \mathcal{E}, \\ & \nabla g_i(\hat{x})d \leq 0, \quad i \in \mathcal{I}_a(\hat{x}). \end{aligned}$$

This has exactly the structure of the direction problem from L12.

The KKT conditions for the linearized problem

Applying the result from L12 to the linearized constraints gives multipliers u_i, $i \in \mathcal{E}$, and $y_i \geq 0$, $i \in \mathcal{I}_a(\hat{x})$, such that

$$\nabla f(\hat{x}) + \sum_{i \in \mathcal{E}} u_i \nabla h_i(\hat{x}) + \sum_{i \in \mathcal{I}_a(\hat{x})} y_i \nabla g_i(\hat{x}) = 0.$$

Setting

$$y_i = 0, \qquad i \in \mathcal{I} \setminus \mathcal{I}_a(\hat{x}),$$

gives

$$\nabla f(\hat{x}) + \sum_{i \in \mathcal{E}} u_i \nabla h_i(\hat{x}) + \sum_{i \in \mathcal{I}} y_i \nabla g_i(\hat{x}) = 0,$$

together with

$$y_i g_i(\hat{x}) = 0, \qquad i \in \mathcal{I}.$$

Why is a constraint qualification needed?

For a linear constraint, the expansion is exact:

$$b - a^{\mathsf{T}}(\hat{x} + td) = b - a^{\mathsf{T}}\hat{x} - ta^{\mathsf{T}}d.$$

For a nonlinear constraint,

$$g_i(\hat{x} + td) = g_i(\hat{x}) + t\nabla g_i(\hat{x})d + o(t).$$

If

$$\nabla g_i(\hat{x})d < 0,$$

then the strict linear decrease dominates the remainder and the constraint is satisfied for positive t sufficiently close to zero.

If

$$\nabla g_i(\hat{x})d = 0,$$

the remainder term determines whether the constraint is satisfied.

A direction feasible for the linearization need not generate feasible points of the nonlinear problem.

A misleading linearization

Consider the constraint

$$g(x) = x^2 \leq 0.$$

The feasible set consists only of

$$x = 0.$$

At $\hat{x} = 0$,

$$g(0) = 0, \qquad g'(0) = 0.$$

The linearized constraint is therefore

$$g(0) + g'(0)d = 0 \leq 0,$$

which is satisfied by every direction $d \in \mathbb{R}$.

However, for every $d \neq 0$ and every $t > 0$, $g(0 + td) = t^2 d^2 > 0$.

Thus no nonzero direction produces nearby feasible points.

The linearization may contain directions that do not represent the true feasible geometry.

Constraint qualifications

A constraint qualification is an assumption ensuring that the linearized constraints describe the local feasible geometry sufficiently well to derive first-order necessary optimality conditions.

In this lecture, we use the Mangasarian–Fromovitz constraint qualification, abbreviated MFCQ.

MFCQ separates two requirements:

- the equality constraints must be locally independent;
- there must be a direction that remains tangent to the equalities and moves strictly into all active inequalities.

Mangasarian–Fromovitz constraint qualification

MFCQ holds at a feasible point $\hat{x}$ if:

1. The equality-constraint gradients

$$\nabla h_i(\hat{x}), \qquad i \in \mathcal{E},$$

   are linearly independent.
2. There exists a vector $\bar{d}$ such that

$$\nabla h_i(\hat{x})\bar{d} = 0, \qquad i \in \mathcal{E},$$

   and

$$\nabla g_i(\hat{x})\bar{d} < 0, \qquad i \in \mathcal{I}_a(\hat{x}).$$

The vector $\bar{d}$ is tangent to the equality constraints and points strictly into every active inequality constraint.

MFCQ rules out a locally degenerate constraint geometry.

Special cases of MFCQ

Only equality constraints.

MFCQ reduces to linear independence of

$$\nabla h_i(\hat{x}), \qquad i \in \mathcal{E}.$$

Equivalently, the Jacobian $\nabla h(\hat{x})$ has full row rank.

Only inequality constraints.

MFCQ requires a direction $\bar{d}$ satisfying

$$\nabla g_i(\hat{x})\bar{d} < 0, \qquad i \in \mathcal{I}_a(\hat{x}).$$

No active inequality constraints.

Only the equality-gradient condition remains.

For linear constraints, the first-order conditions from L12 hold without imposing MFCQ separately.

What MFCQ provides

Suppose that $\hat{x}$ is feasible and MFCQ holds.

If a vector d satisfies

$$\nabla f(\hat{x})d < 0,$$

$$\nabla h_i(\hat{x})d = 0, \qquad i \in \mathcal{E},$$

and

$$\nabla g_i(\hat{x})d \leq 0, \qquad i \in \mathcal{I}_a(\hat{x}),$$

then $\hat{x}$ cannot be a local minimizer.

The reason is that MFCQ allows the linearized direction to be modified, if necessary, into a nearby direction that

- remains tangent to the equality constraints;
- points strictly into all active inequalities;
- still decreases the objective.

A nearby feasible curve can then be constructed with this initial direction.

First-order necessary optimality conditions

Theorem (KKT conditions)

Assume that $\hat{x}$ is a local minimizer of (P) and that MFCQ holds at $\hat{x}$. Then there exist multipliers $\hat{u}_i$, $i \in \mathcal{E}$, and $\hat{y}_i$, $i \in \mathcal{I}$, satisfying the following conditions.

1. $\nabla f(\hat{x}) + \sum_{i \in \mathcal{E}} \hat{u}_i \nabla h_i(\hat{x}) + \sum_{i \in \mathcal{I}} \hat{y}_i \nabla g_i(\hat{x}) = 0.$ *(Stationarity)*
2. $h_i(\hat{x}) = 0, \quad i \in \mathcal{E}, \qquad g_i(\hat{x}) \leq 0, \quad i \in \mathcal{I}.$ *(Feasibility)*
3. $\hat{y}_i \geq 0, \qquad i \in \mathcal{I}.$ *(Nonnegativity of inequality multipliers)*
4. $\hat{y}_i g_i(\hat{x}) = 0, \qquad i \in \mathcal{I}.$ *(Complementarity)*

Equality multipliers $\hat{u}_i$ are free. Inequality multipliers $\hat{y}_i$ are nonnegative.

Only active inequality constraints can have nonzero multipliers.

The Lagrangian

Define the Lagrangian

$$L(x, u, y) = f(x) + \sum_{i \in \mathcal{E}} u_i h_i(x) + \sum_{i \in \mathcal{I}} y_i g_i(x).$$

Its gradient with respect to x is

$$\nabla_x L(x, u, y) = \nabla f(x) + \sum_{i \in \mathcal{E}} u_i \nabla h_i(x) + \sum_{i \in \mathcal{I}} y_i \nabla g_i(x).$$

Thus KKT stationarity is

$$\nabla_x L(\hat{x}, \hat{u}, \hat{y}) = 0.$$

The complete KKT conditions are:

- stationarity;
- feasibility;
- nonnegativity of inequality multipliers;
- complementarity.

A geometric interpretation

At a KKT point,

$$-\nabla f(\hat{x})^{\mathsf{T}} = \sum_{i \in \mathcal{E}} \hat{u}_i \nabla h_i(\hat{x})^{\mathsf{T}} + \sum_{i \in \mathcal{I}_a(\hat{x})} \hat{y}_i \nabla g_i(\hat{x})^{\mathsf{T}}.$$

Thus the negative objective gradient is represented by normals to the active constraint surfaces.

- Equality normals may be used with either sign.
- Active inequality normals may only be used with nonnegative coefficients.
- Inactive inequality normals do not appear.
- Stationarity may be interpreted as a force balance between the objective and the active constraints.

At a local minimizer, the objective cannot decrease along the local feasible geometry.

A nonlinear equality example

Consider

$$\begin{aligned} &\text{minimize} && \tfrac{1}{2}\left(x_1^2 + x_2^2\right) \\ &\text{subject to} && h(x) = x_1 x_2 - 1 = 0. \end{aligned}$$

The Lagrangian is

$$L(x, u) = \frac{1}{2}\left(x_1^2 + x_2^2\right) + u(x_1 x_2 - 1).$$

The KKT conditions are

$$\begin{aligned} x_1 + u x_2 &= 0, \\ x_2 + u x_1 &= 0, \\ x_1 x_2 &= 1. \end{aligned}$$

The feasible points closest to the origin are

$$\hat{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \hat{x} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \quad \text{with} \quad \hat{u} = -1.$$

A nonlinear inequality example

Consider

$$\begin{aligned} &\text{minimize} && \tfrac{1}{2}\left((x_1-2)^2+x_2^2\right) \\ &\text{subject to} && g(x)=x_1^2+x_2^2-1\le 0. \end{aligned}$$

The unconstrained minimizer $(2,0)$ is infeasible, so the constraint is active at the constrained minimizer.

The KKT conditions are

$$\begin{pmatrix}\hat{x}_1-2 & \hat{x}_2\end{pmatrix}+\hat{y}\begin{pmatrix}2\hat{x}_1 & 2\hat{x}_2\end{pmatrix}=0,$$

$$\hat{x}_1^2+\hat{x}_2^2-1=0, \qquad \hat{y}\ge 0.$$

They give

$$\hat{x}=\begin{pmatrix}1\\0\end{pmatrix}, \qquad \hat{y}=\frac{1}{2}.$$

The active constraint normal balances the objective gradient.

When MFCQ fails

Return to

$$\begin{array}{ll} \text{minimize} & x \\ \text{subject to} & x^2 \leq 0. \end{array}$$

The only feasible point is $\hat{x} = 0$, so $\hat{x}$ is a global minimizer.

However,

$$g'(0) = 0,$$

so MFCQ fails: there is no direction d satisfying

$$g'(0)d < 0.$$

KKT stationarity would require

$$f'(0) + \hat{y}g'(0) = 1 + \hat{y}\,0 = 0,$$

which is impossible.

Without a constraint qualification, a local minimizer need not admit KKT multipliers.

Convex problems: KKT conditions are sufficient

Assume that

- f is convex;
- each g_i is convex;
- each h_i is affine.

Suppose that $\hat{x}$, $\hat{u}$, and $\hat{y}$ satisfy the KKT conditions. Then the Lagrangian

$$L(x, \hat{u}, \hat{y}) = f(x) + \sum_{i \in \mathcal{E}} \hat{u}_i h_i(x) + \sum_{i \in \mathcal{I}} \hat{y}_i g_i(x)$$

is convex as a function of x.

Stationarity gives

$$\nabla_x L(\hat{x}, \hat{u}, \hat{y}) = 0,$$

so $\hat{x}$ globally minimizes the Lagrangian.

Primal feasibility and complementarity then imply that $\hat{x}$ is a global minimizer of the original problem.

Necessary and sufficient conditions

For a general nonlinear problem:

$$\text{local minimizer} + \text{MFCQ} \quad \Longrightarrow \quad \text{KKT conditions.}$$

Thus KKT conditions are first-order necessary conditions under a constraint qualification.

For a convex problem with convex inequalities and affine equalities:

$$\text{KKT conditions} \Longrightarrow \text{global minimizer.}$$

Thus KKT conditions are also sufficient for global optimality in the convex case.

Regularity gives necessity. Convexity gives sufficiency.

Main ideas

After this lecture, you should be able to:

- linearize equality and inequality constraints at a feasible point;
- formulate the linearized feasible-direction problem;
- explain why a constraint qualification is needed;
- state and interpret MFCQ;
- state the KKT conditions for nonlinear constraints;
- express stationarity using the Lagrangian;
- explain what may happen when MFCQ fails;
- explain why KKT conditions are sufficient for global optimality in a convex problem.

Under MFCQ, the linearized feasible geometry yields the first-order optimality conditions of the nonlinear problem.